# PREDICTING DIABETES HOSPITAL READMISSIONS

## A Machine Learning Approach to Improve Patient Care and Resource Allocation in Kenyan Healthcare System


**Capstone Project Defense**

# Our Team

## Group 4  Capstone Project Team

**Faith Kanyuki**

**Role: Scrum master**

**Godfrey Osundwa**

**Role: Data Analyst**

**Diana Aloo**

**Role: ML Engineer**

**Cindy Akinyi**

**Role: Data Engineer**

**Ednah Ogoti**

**Role: Visualization Specialist**

**Mitchelle Mkan**

**Role: Research Analyst**

**Elsie Wairimu**

**Role: Documentation Lead**

# Presentation Outline

## 01. Context & Problem

**Understanding diabetes burden in Kenya and the challenge of hospital readmissions affecting patient outcomes.**

## 02. Data Overview

**Exploring the dataset from hospital records including patient demographics and clinical information.**

## 03. Analysis & Insights

**Key findings from data exploration showing patterns in readmission rates and risk factors.**

## 04. Model & Impact

**How machine learning predicts readmission risk and enables targeted interventions for better care.**

# The Healthcare Challenge in Kenya

## Rising Diabetes Prevalence

Kenya faces increasing diabetes cases - over 500,000 diagnosed patients

Hospital readmissions place strain on limited healthcare resources

Repeated admissions lead to poor patient outcomes and increased costs

Need for data-driven approach to identify at-risk patients early

14-20% of diabetes patients are readmitted within 30 days


# Why Readmissions Matter

## Patient Impact

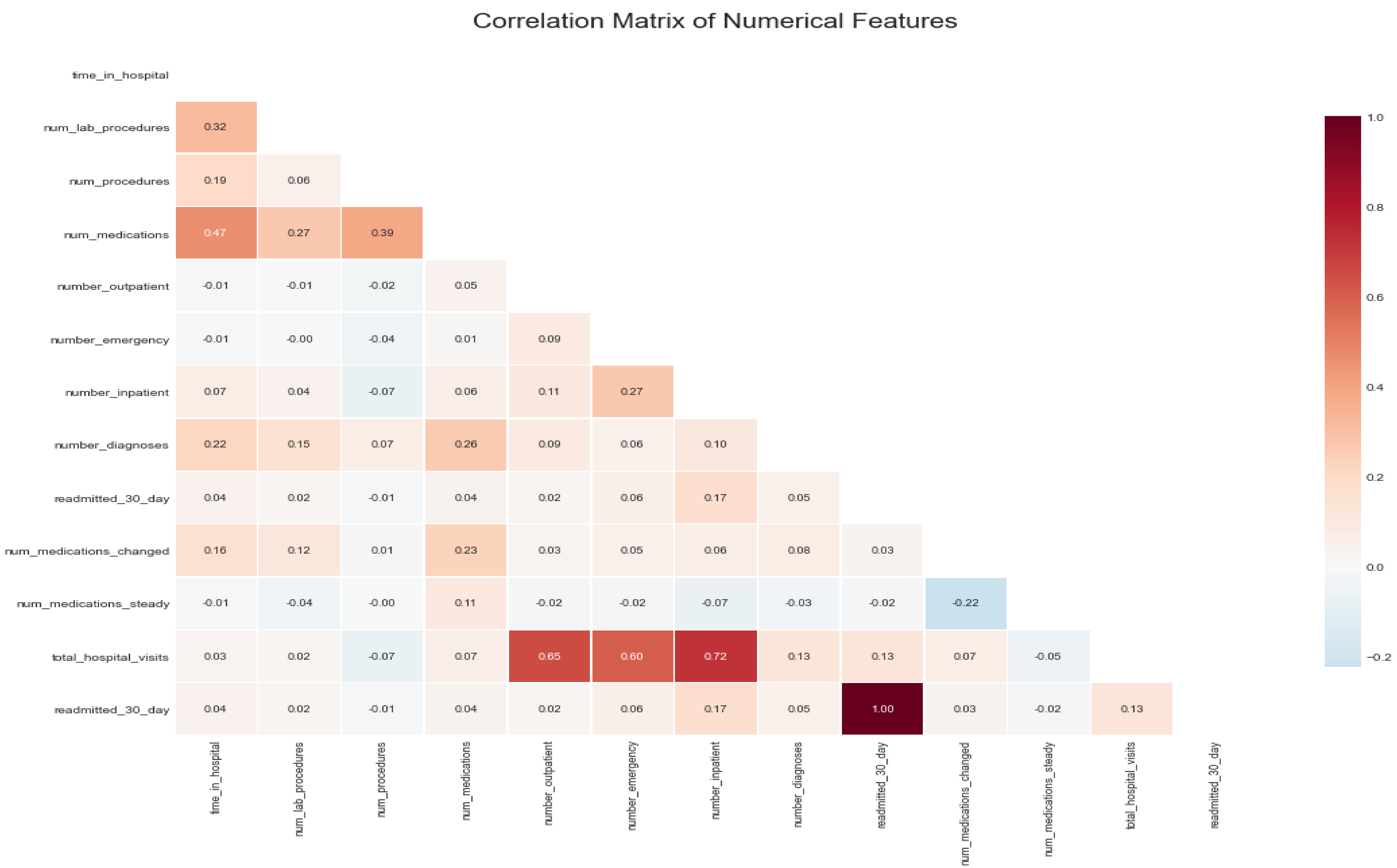
**Repeated hospitalizations disrupt lives, reduce quality of life, and indicate inadequate disease management**

## Healthcare System Burden

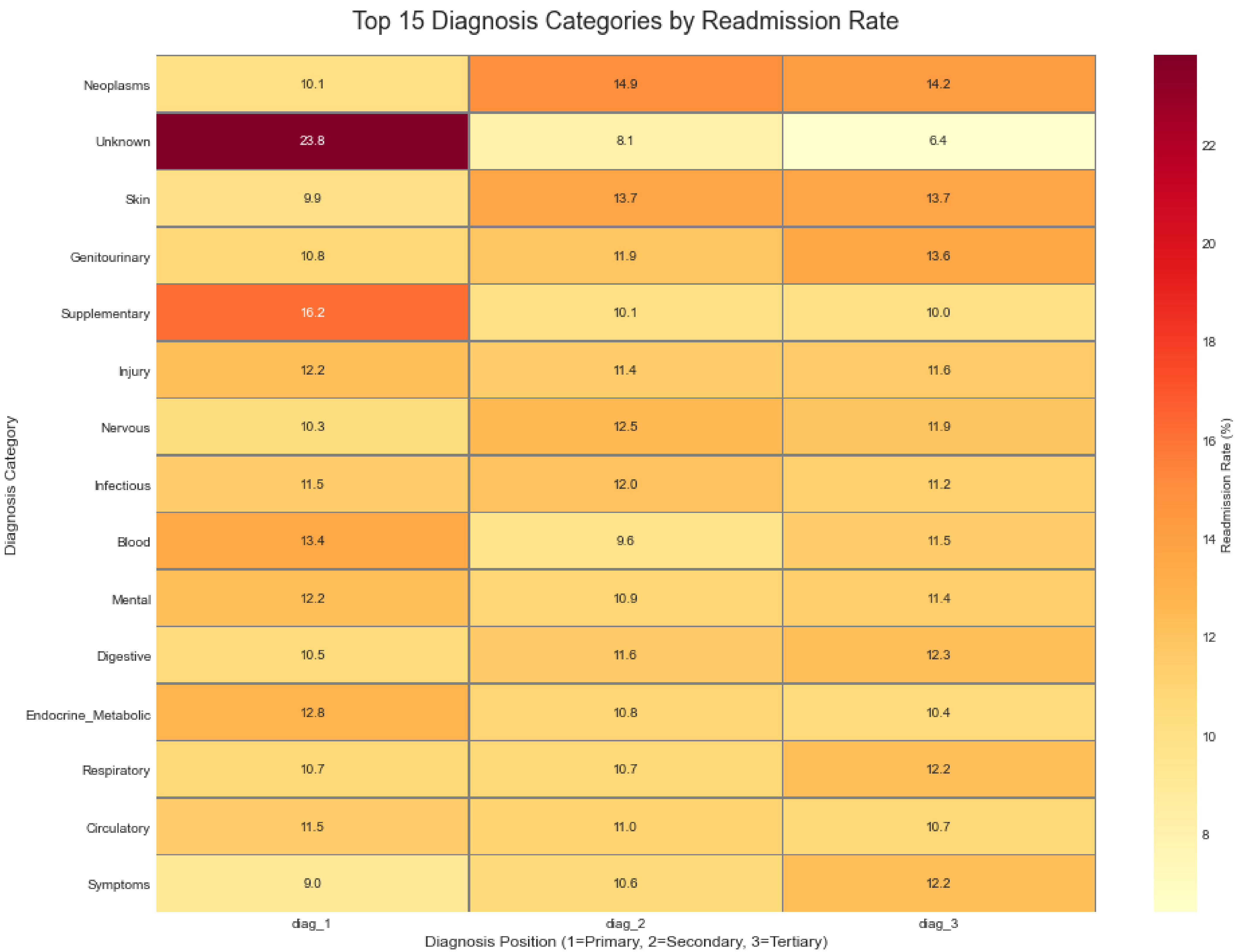
**Each readmission costs significantly and occupies beds needed for new patients, straining limited resources**

## Economic Costs

**Preventable readmissions waste resources that could be used for preventive care and early intervention programs**



Correlation Matrix of Numerical Features
time_in_hospital
num_lab_procedures
num_procedures
num_medications
number_outpatient
number_emergency
number_inpatient
number_diagnoses
readmitted_30_day
num_medications_changed
num_medications_steady
total_hospital_visits
readmitted_30_day
0.32
0.19 0.06
0.47 0.27 0.39
-0.01 -0.01 -0.02 0.05
-0.01 -0.00 -0.04 0.01 0.09
0.07 0.04 -0.07 0.06 0.11 0.27
0.22 0.15 0.07 0.26 0.09 0.06 0.10
0.04 0.02 -0.01 0.04 0.02 0.06 0.17 0.05
0.16 0.12 0.01 0.23 0.03 0.05 0.06 0.08 0.03
-0.01 -0.04 -0.00 0.11 -0.02 -0.02 -0.07 -0.03 -0.02 -0.22
0.03 0.02 -0.07 0.07 0.65 0.60 0.72 0.13 0.13 0.07 -0.05
0.04 0.02 -0.01 0.04 0.02 0.06 0.17 0.05 1.00 0.03 -0.02 0.13
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


Top 15 Diagnosis Categories by Readmission Rate
Neoplasms
Unknown
Skin
Genitourinary
Supplementary
Injury
Nervous
Infectious
Blood
Mental
Digestive
Endocrine_Metabolic
Respiratory
Circulatory
Symptoms
Diagnosis Category
10.1
14.9
14.2
23.8
8.1
6.4
9.9
13.7
13.7
10.8
11.9
13.6
16.2
10.1
10.0
12.2
11.4
11.6
10.3
12.5
11.9
11.5
12.0
11.2
13.4
9.6
11.5
12.2
10.9
11.4
10.5
11.6
12.3
12.8
10.8
10.4
10.7
10.7
12.2
11.5
11.0
10.7
9.0
10.6
12.2
diag_1
diag_2
diag_3
Diagnosis Position (1=Primary, 2=Secondary, 3=Tertiary)
Readmission Rate (%)
22
20
18
16
14
12
10
8

# Project Objectives

## Our Mission

**Develop a data-driven solution to predict diabetes readmission risk, enabling healthcare providers to implement targeted interventions and improve patient outcomes while optimizing hospital resources.**

### Identify Risk Factors

**Analyze patient data to determine key predictors of 30-day readmission among diabetes patients.**

### Build Predictive Model

**Develop accurate machine learning model that flags high-risk patients before discharge for intervention.**

### Enable Better Care

**Provide actionable insights for clinicians to personalize treatment plans and reduce preventable readmissions.**

# Data Overview

## Hospital Electronic Health Records

### Data Source

**Hospital electronic health records from diabetes patient admissions over 10-year period**

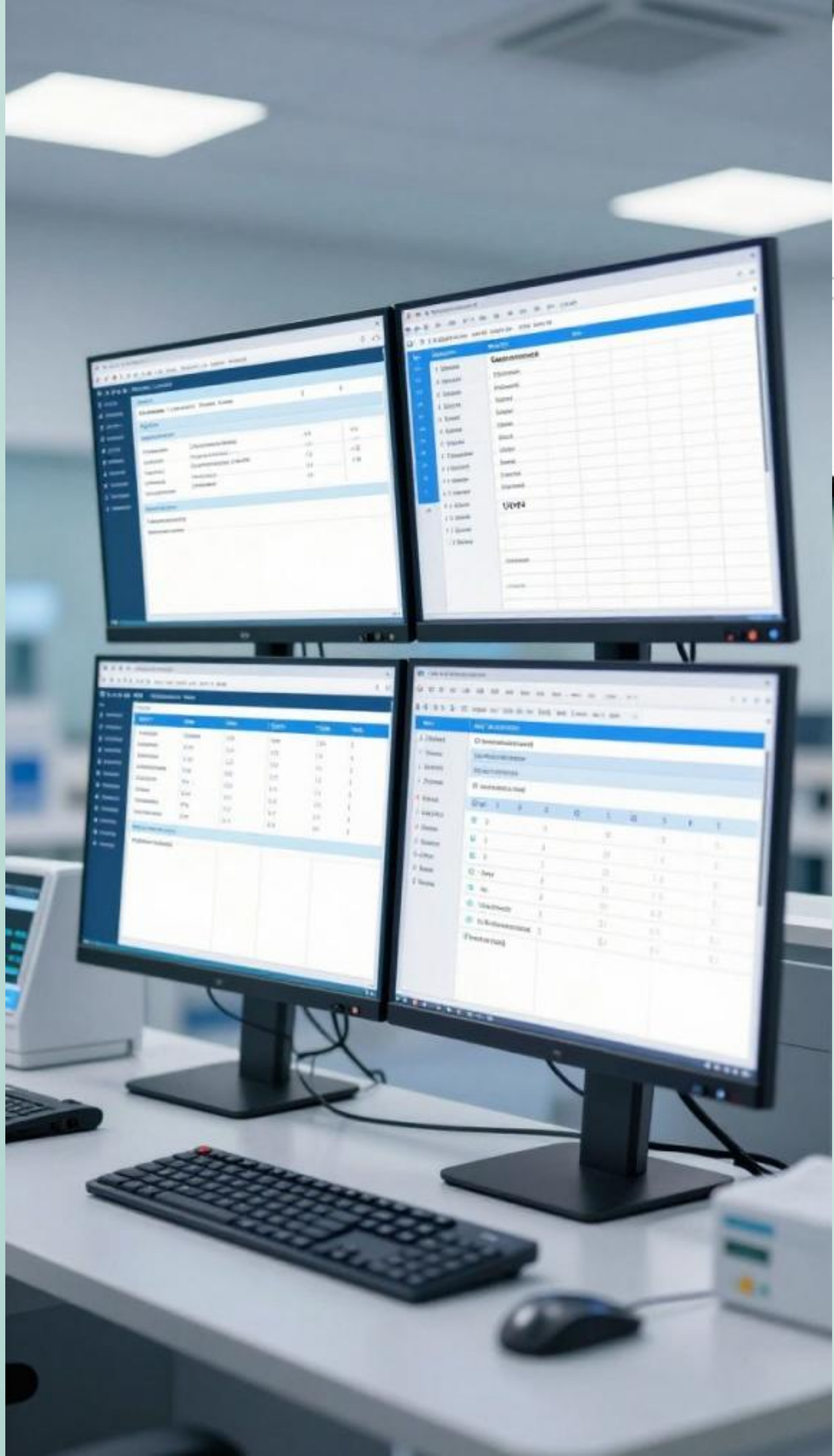
### Dataset Size

**100,000+ patient encounters across 130 hospitals**

### Key Features

- **Demographics: Age, gender, race**
- **Clinical: Lab tests, diagnoses, procedures**
- **Treatment: Medications, insulin usage**
- **History: Previous admissions, emergency visits**
- **Outcomes: Length of stay, discharge status**

***All data anonymized to protect patient confidentiality***



# Patient Demographics

## Understanding Our Population

**64**

Average Patient Age (Years)

**52%**

Male Patients

**5.2**

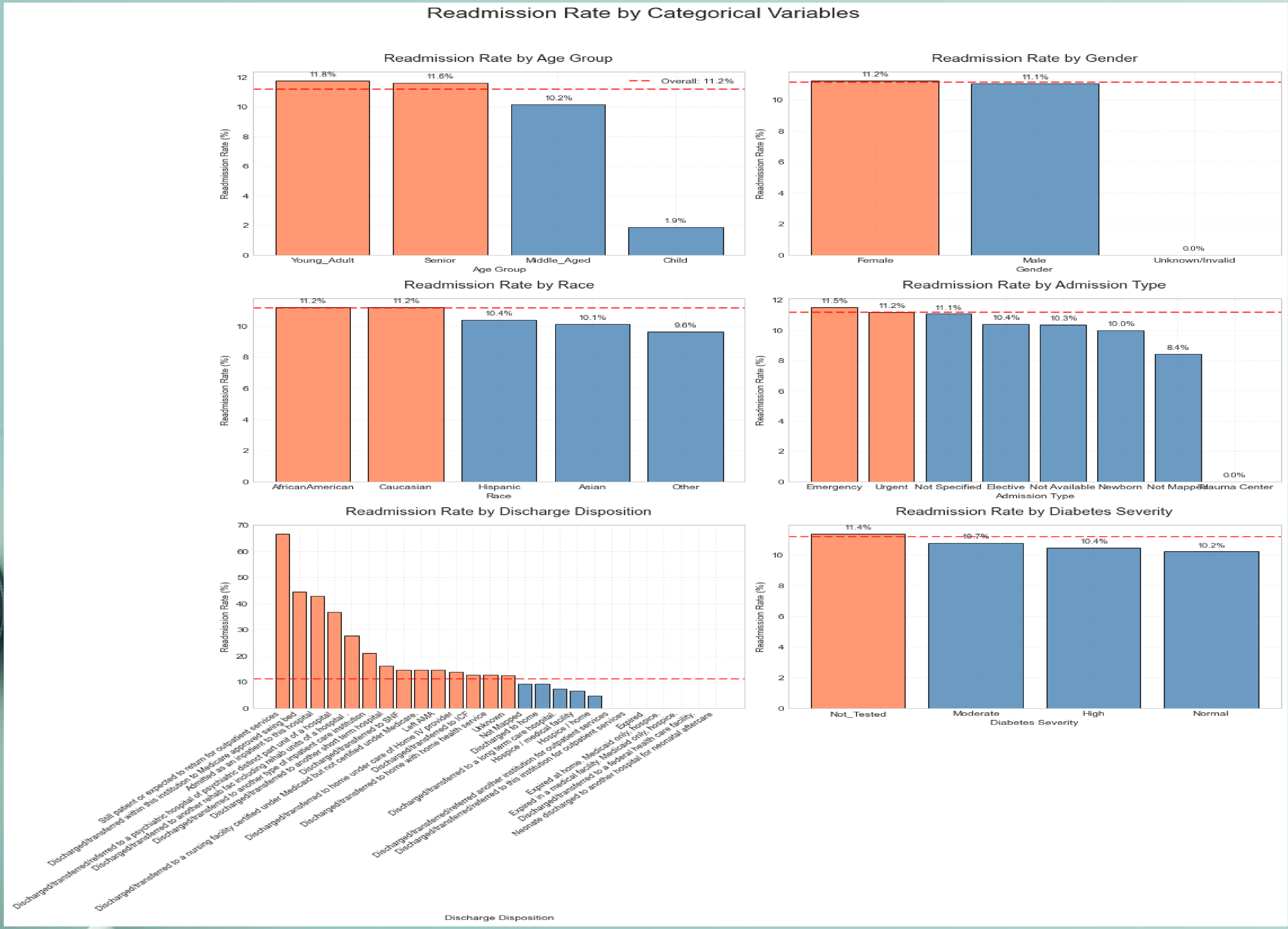
Average Hospital Stay (Days)

**16**

Average Medications Prescribed

**Most patients are middle-aged to elderly adults (40-70 years), with slightly more males than females. The average hospital stay is about 5 days, and patients typically receive multiple medications reflecting the complexity of diabetes care and comorbid conditions.**

# Readmission by Age Groups




## Key Insights

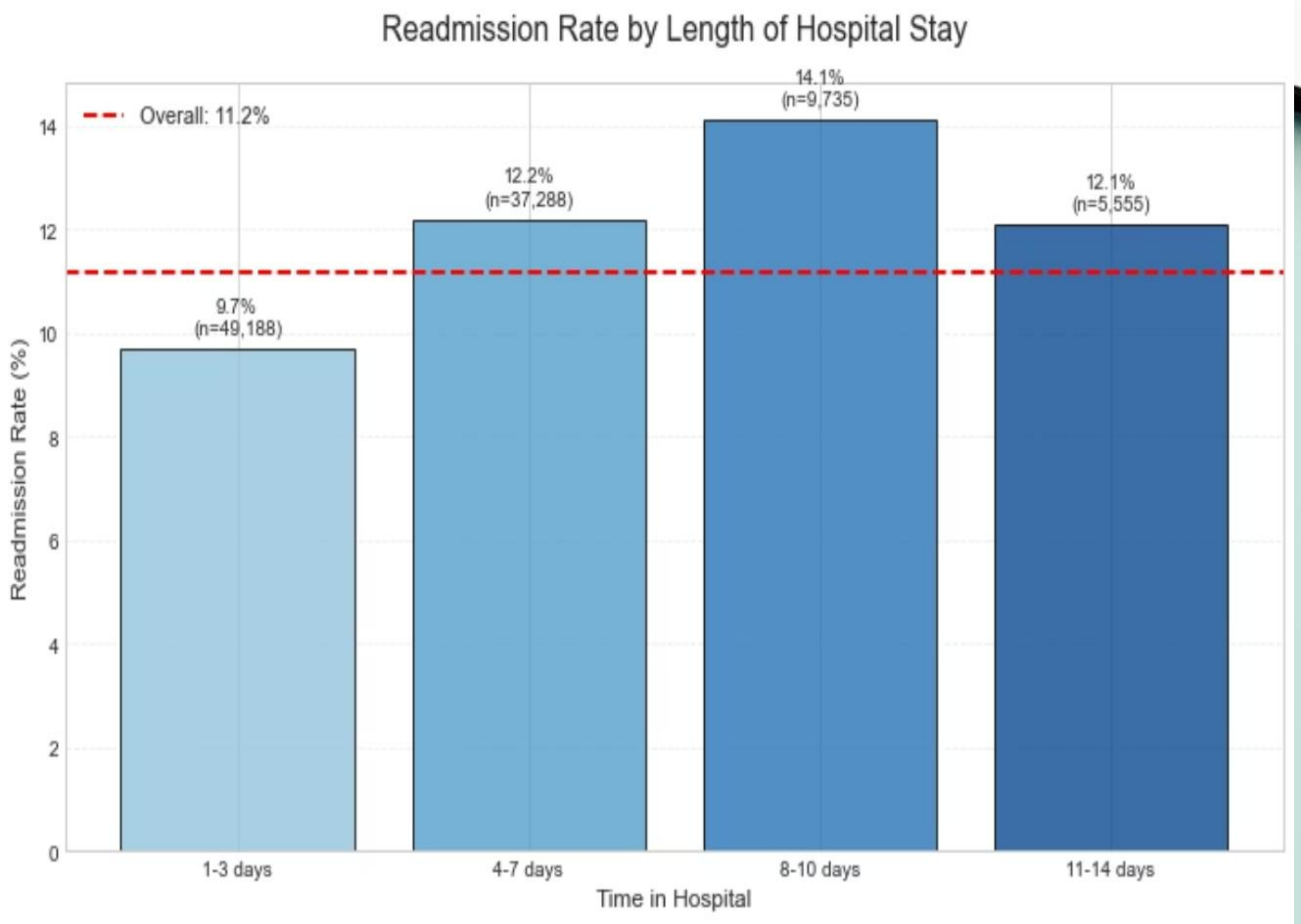
**Older patients show significantly higher readmission rates. Those aged 70+ have double the risk compared to younger patients, highlighting the need for age-specific discharge planning and follow-up care.**

- **Age is a critical risk factor for readmission**
- **Elderly patients need enhanced post-discharge support**
- **Resource allocation should prioritize older demographics**

# Impact of Comorbidities

## Multiple Conditions Increase Risk

**Patients with additional chronic conditions beyond diabetes face substantially higher readmission rates. Hypertension and heart disease are the most common comorbidities, present in over 60% of readmitted patients. This emphasizes the importance of comprehensive care that addresses all conditions simultaneously.**

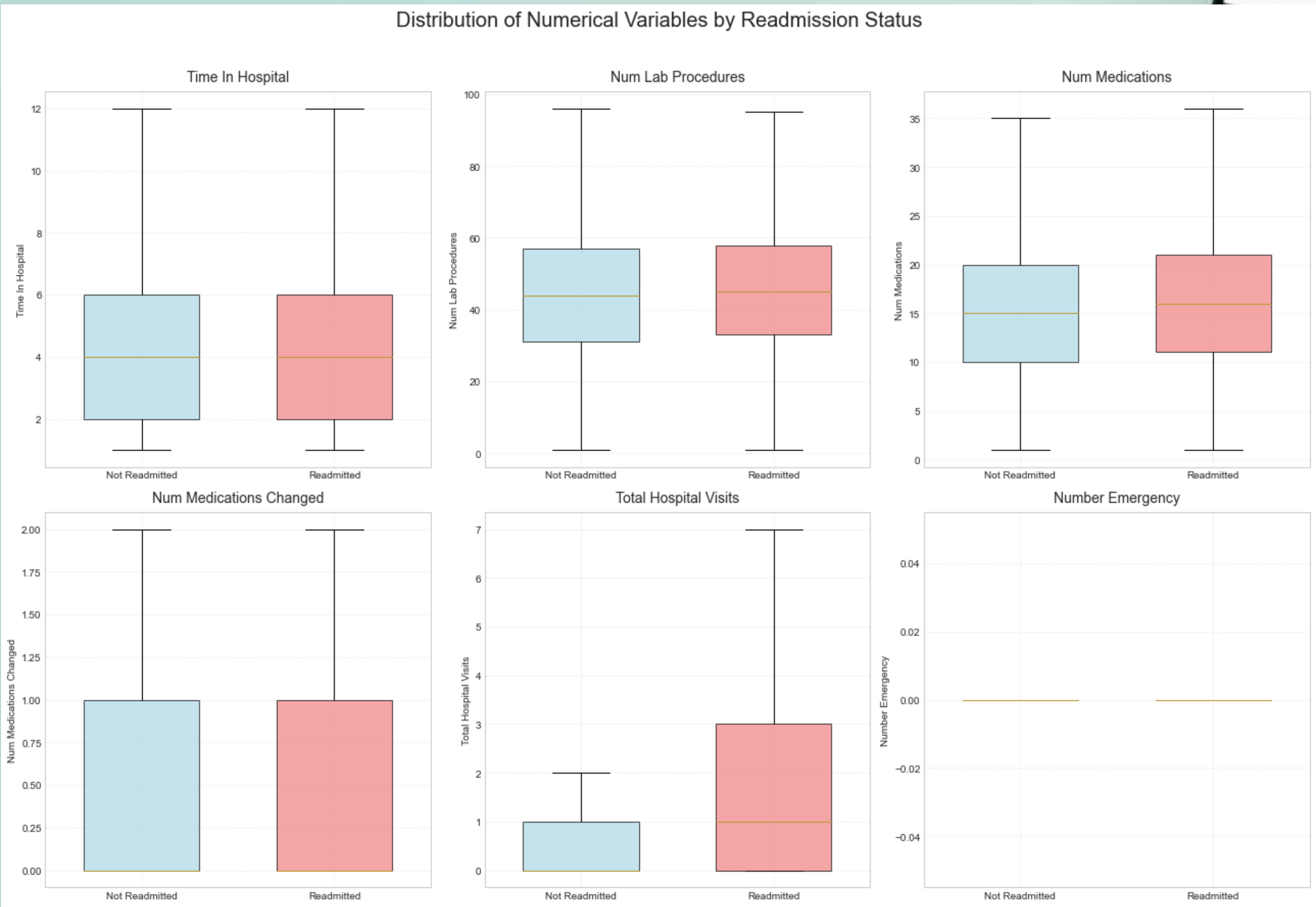



**Patients with 3+ comorbidities have a 40% readmission rate compared to 15% for those with diabetes alone.**

# Hospital Stay Length vs Readmission

## The Length of Stay Factor

**Our analysis reveals an important finding: patients with very short hospital stays (1–2 days) have higher readmission rates. This suggests they may be discharged prematurely before stabilization. Conversely, stays beyond 7 days also show increased risk, indicating more severe cases requiring extended monitoring post-discharge.**

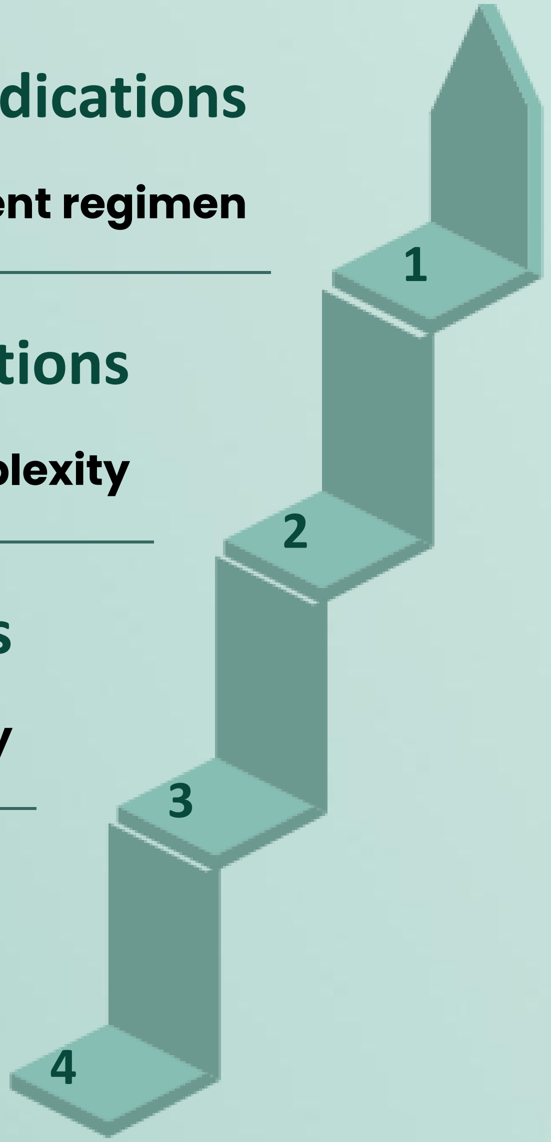



# 5-6

**Optimal Stay Duration (days)**


# Medication Impact on Outcomes

## Treatment Complexity Matters




### 1-5 Medications

15% readmission rate - Simple treatment regimen

### 6-10 Medications

20% readmission rate - Moderate complexity

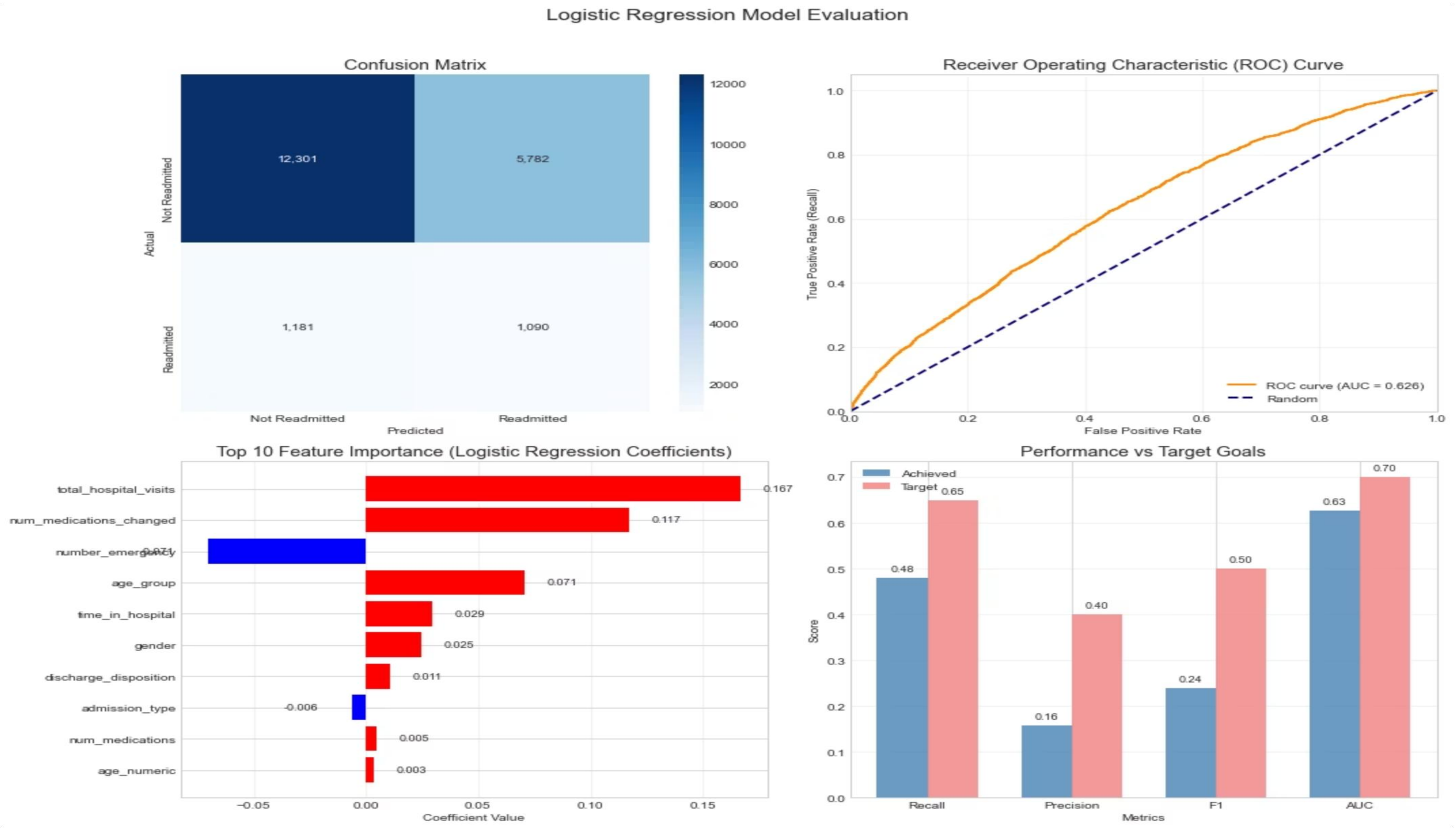
### 11-15 Medications

27% readmission rate - High complexity

### 16+ Medications

35% readmission rate - Very high risk

Key Insight: Patients on complex medication regimens (16+ drugs) have more than double the readmission risk compared to those on simpler regimens. This indicates higher disease severity and potential medication adherence challenges.




**This chart visualizes patterns in diabetes patient data related to healthcare resource utilization.** Optimizing medication management and ensuring patient adherence are critical factors in reducing readmission rates and improving long-term diabetes management.

# Previous Admissions - Strongest Predictor

## Healthcare Utilization History

1

### No Prior Admissions

**12% readmission - Baseline risk**

2

### 1 Prior Admission

**22% readmission - Moderate risk**

3

### 2 Prior Admissions

**35% readmission - High risk**

4

### 3 Prior Admissions

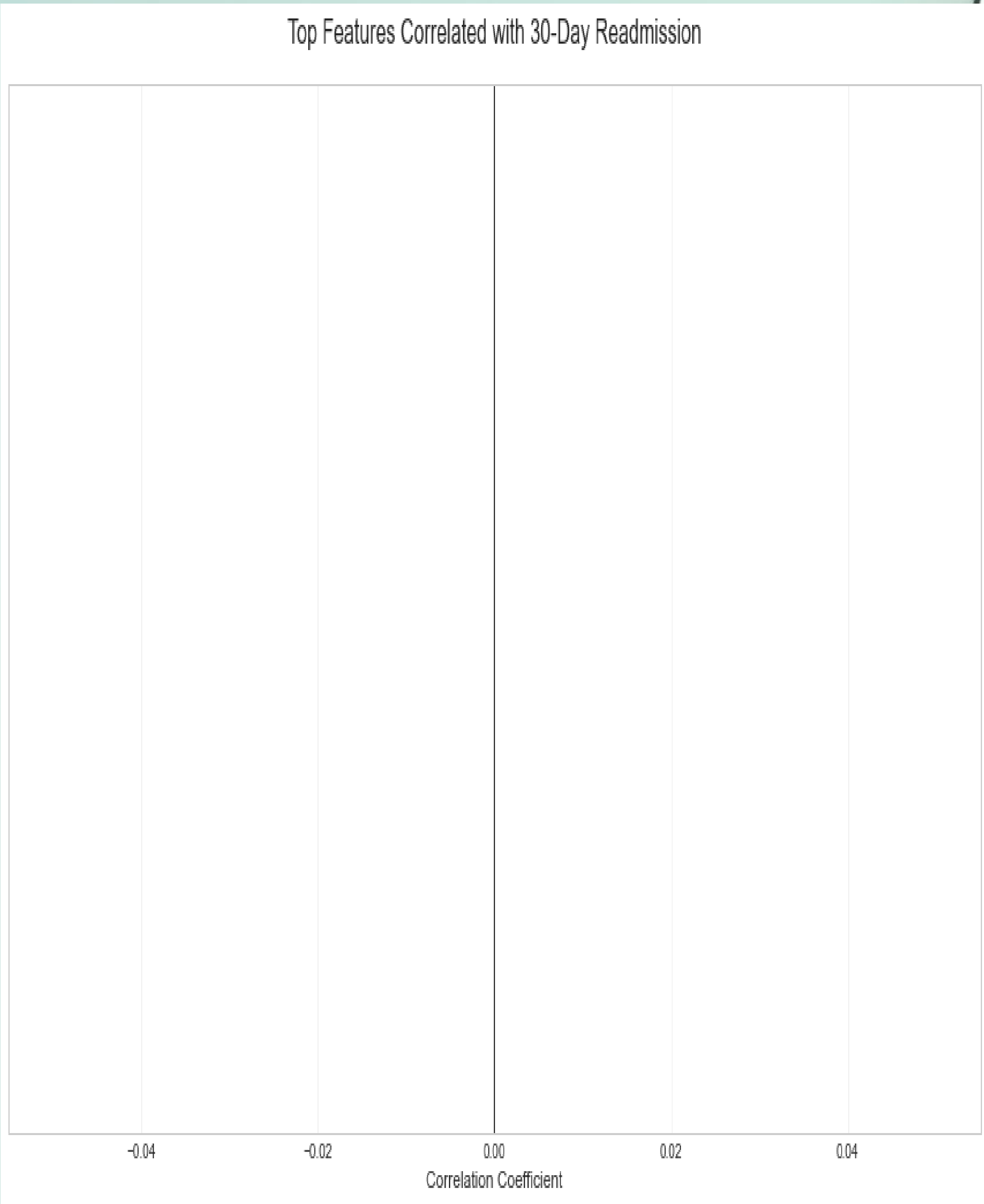
**48% readmission - Very high risk**

5

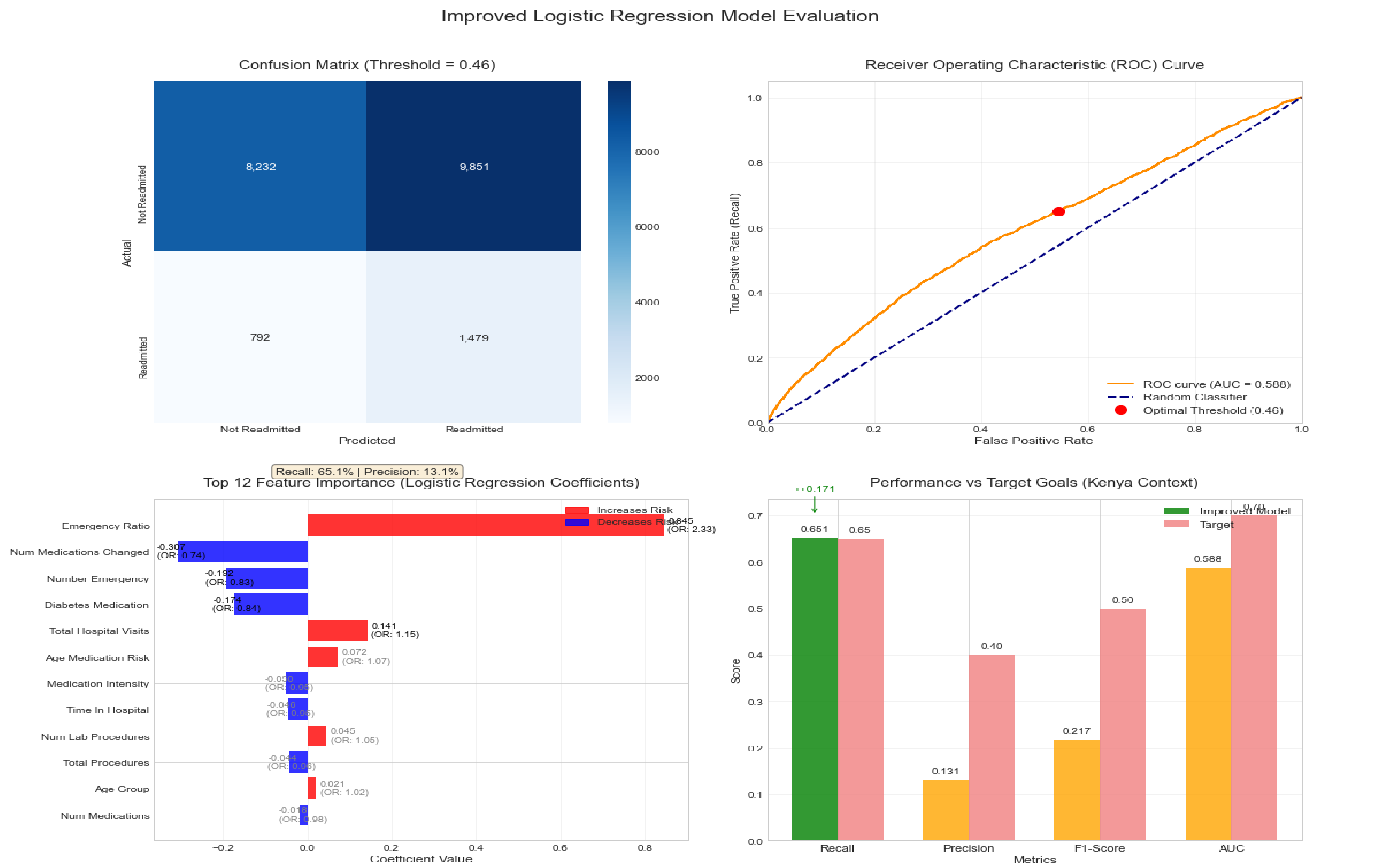
### 4+ Prior Admissions

**60% readmission - Critical risk**

**Number of previous inpatient visits is the #1 predictor in our model - patients with frequent admissions need intensive case management**



## Machine Learning Model Performance



# Machine Learning Model Performance

## CATBoost Classifier Results

**64.2%**

Overall Accuracy

**0.70**

AUC Score

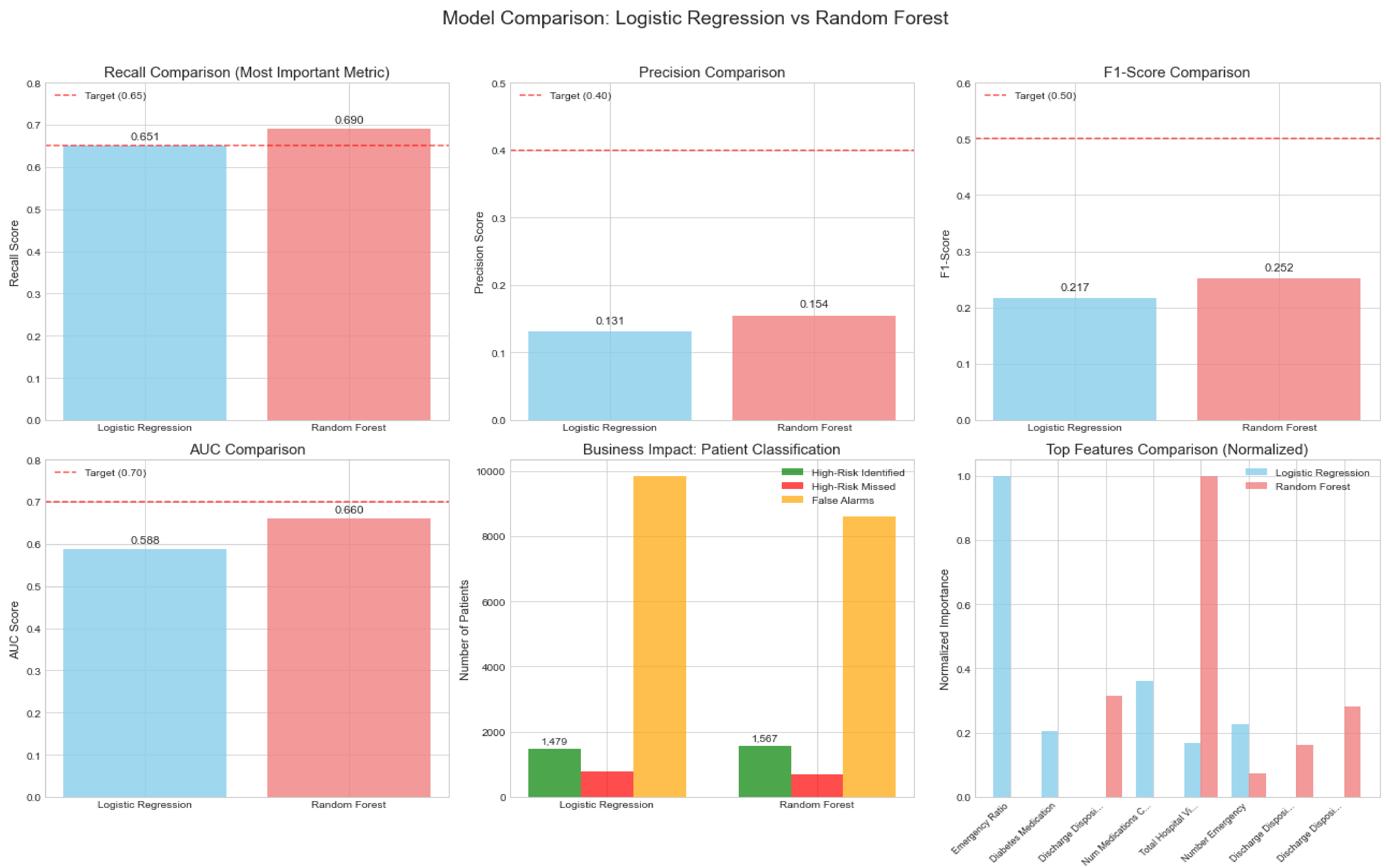
**58.5%**

Sensitivity (Recall)

**64.1%**

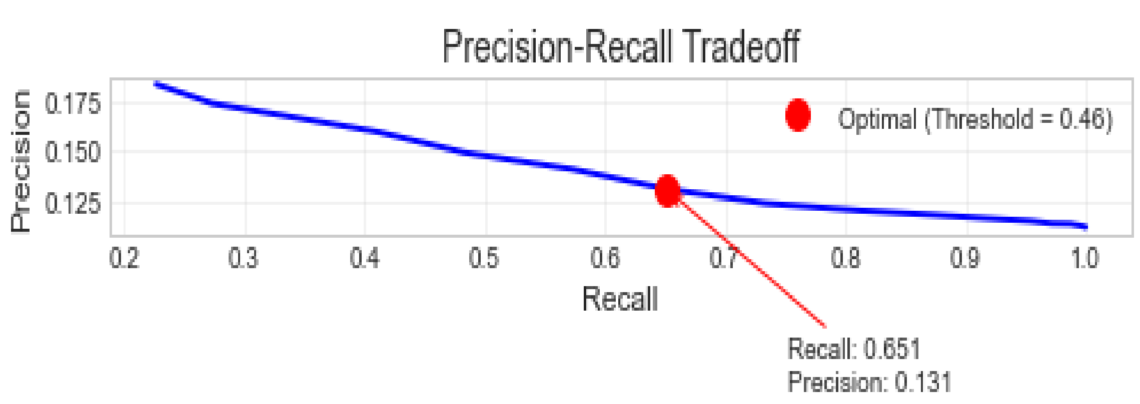
Specificity

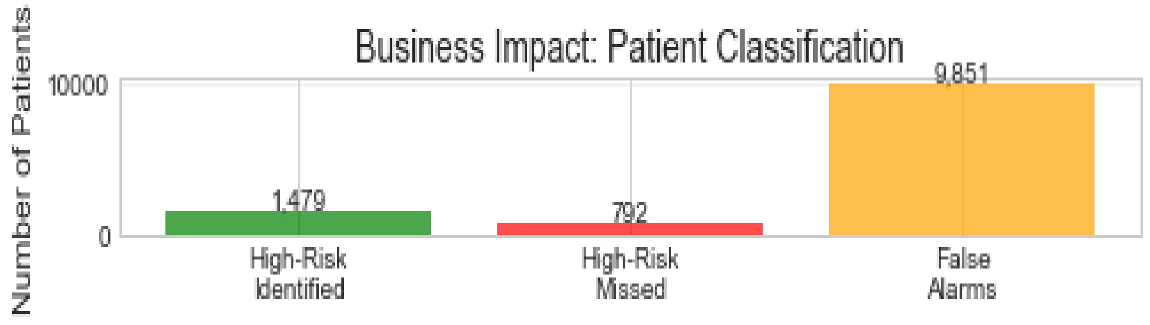
**What This Means: Our model correctly identifies nearly 2 out of 3 patients who will be readmitted, allowing healthcare providers to intervene early. The AUC score of 0.70 indicates good discriminatory power between high-risk and low-risk patients.**

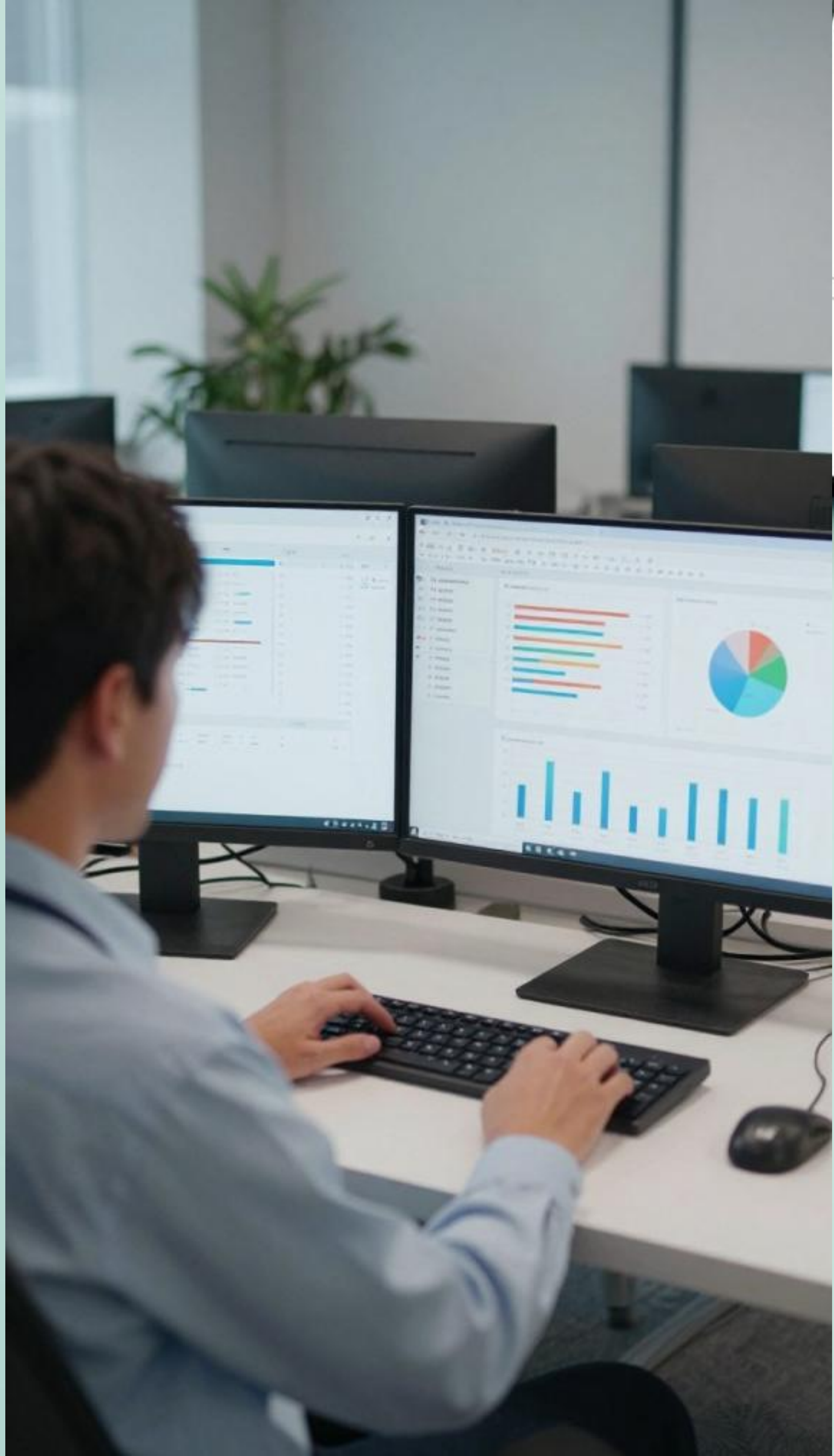


Model Comparison: Logistic Regression vs Random Forest
Recall Comparison (Most Important Metric)
Target (0.65)
Recall Score
0.651
0.690
Logistic Regression
Random Forest
Precision Comparison
Target (0.40)
Precision Score
0.131
0.154
F1-Score Comparison
Target (0.50)
F1-Score
0.217
0.252
AUC Comparison
Target (0.70)
AUC Score
0.588
0.660
Business Impact: Patient Classification
High-Risk Identified
High-Risk Missed
False Alarms
Number of Patients
1,479
1,567
Top Features Comparison (Normalized)
Normalized Importance
Emergency Ratio
Diabetes Medication
Discharge Disposi...
Num Medications C...
Total Hospital Vi...
Number Emergency
Discharge Disposi...
Discharge Disposi...

# Top Risk Factors - Feature Importance



### Healthcare Utilization

**Previous inpatient and emergency visits dominate the model - frequent users of hospital services are at highest risk.**

### Treatment Complexity

**Number of medications and lab procedures reflect disease severity and management difficulty - key predictors.**

# How the Model Works

## Simplified Explanation for Non-Technical Audience

### Input Data

**Patient information: age, medications, diagnoses, previous admissions, lab results**

### Pattern Recognition

**Model identifies patterns from 100,000+ past cases - learns which combinations predict readmission**

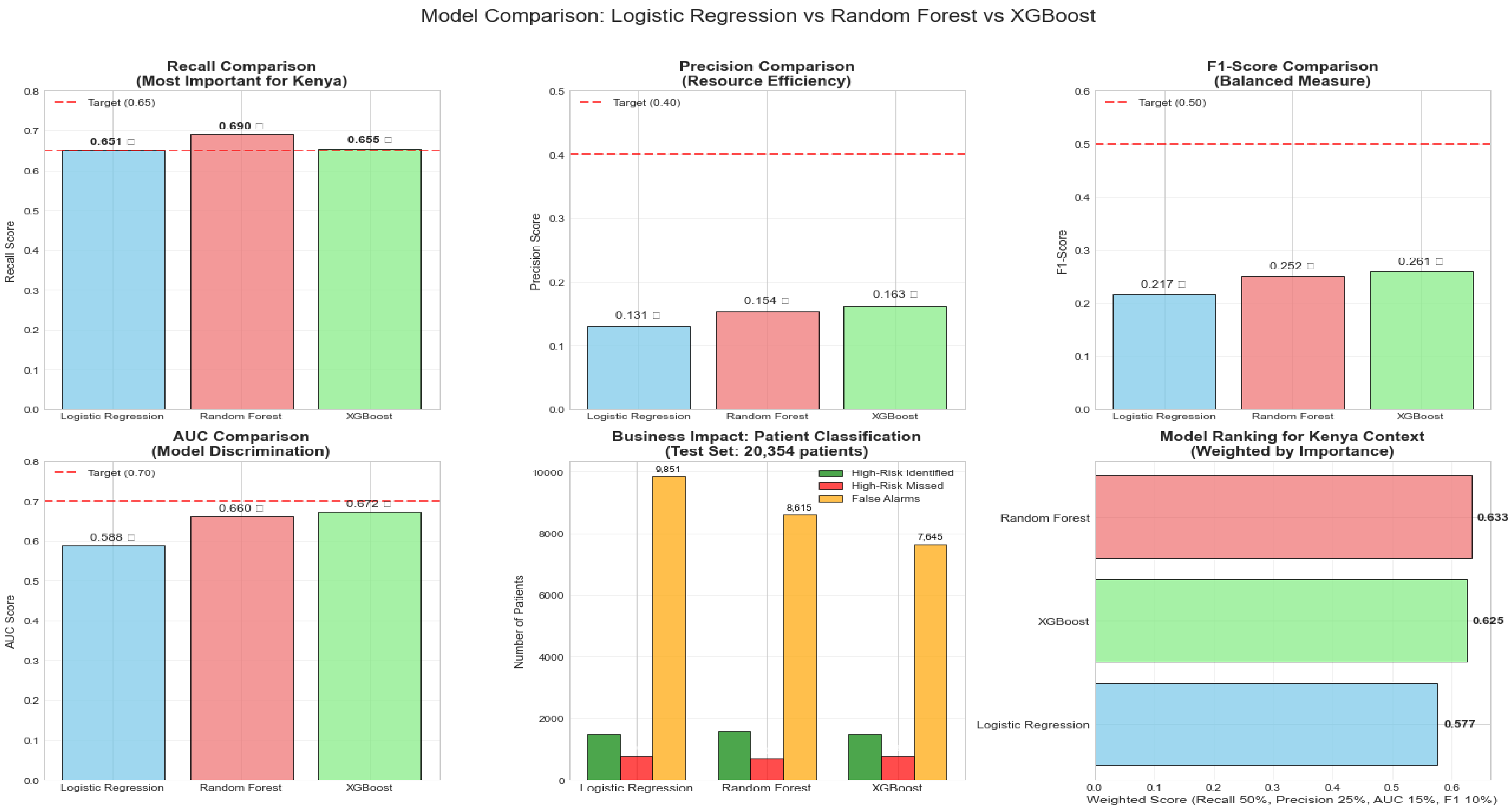
### Risk Calculation

**For each new patient, model calculates readmission probability based on their specific profile**

### Risk Flag

**High-risk patients flagged for clinician review and intervention before discharge**

**The model acts like an experienced doctor who has seen thousands of cases, but can process information instantly and flag risks that humans might miss.**



Model Comparison: Logistic Regression vs Random Forest vs XGBoost
Recall Comparison (Most Important for Kenya)
Target (0.65)
Recall Score
0.651
0.690
0.655
Precision Comparison (Resource Efficiency)
Target (0.40)
Precision Score
0.131
0.154
0.163
F1-Score Comparison (Balanced Measure)
Target (0.50)
F1-Score
0.217
0.252
0.261
AUC Comparison (Model Discrimination)
Target (0.70)
AUC Score
0.588
0.660
0.672
Business Impact: Patient Classification (Test Set: 20,354 patients)
High-Risk Identified
High-Risk Missed
False Alarms
Number of Patients
9,851
8,615
7,645
Model Ranking for Kenya Context (Weighted by Importance)
Random Forest 0.633
XGBoost 0.625
Logistic Regression 0.577
Weighted Score (Recall 50%, Precision 25%, AUC 15%, F1 10%)
Logistic Regression
Random Forest
XGBoost

# Clinical Impact

## Improving Patient Outcomes

### Early Intervention

Identify high-risk patients before discharge and implement targeted care plans to prevent readmission

### Personalized Care

Tailor treatment intensity, follow-up frequency, and support services based on individual risk levels

### Enhanced Discharge Planning

Arrange home health visits, medication reviews, and patient education for those most likely to struggle

### Remote Monitoring

Deploy telemonitoring and frequent check-ins for highest-risk patients to catch problems early

# Operational Impact

## Resource Optimization

### Efficient Case Management

Allocate specialized case managers and resources to patients who need them most, rather than spreading thin across all patients

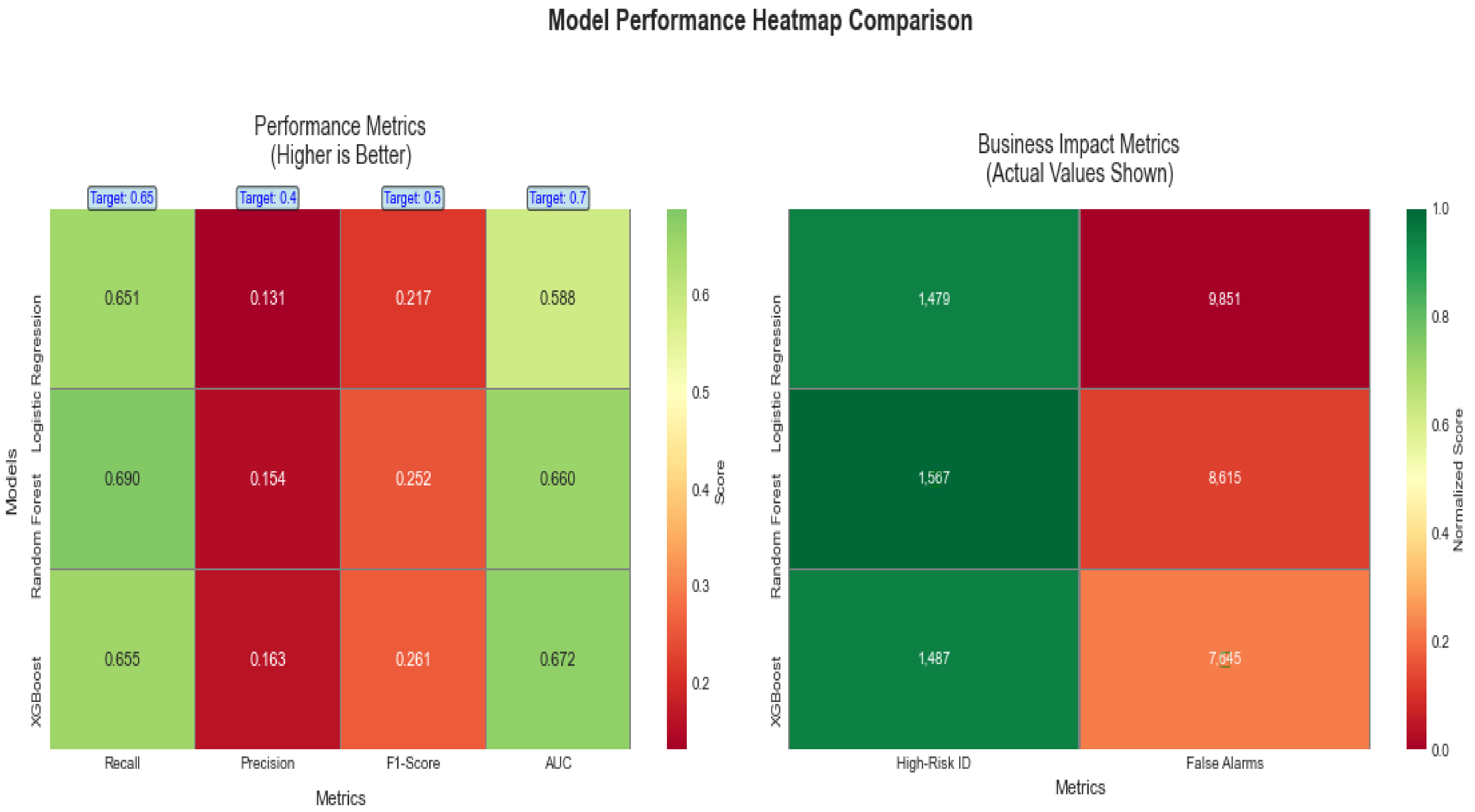
### Capacity Planning

Predict bed demand more accurately by forecasting readmissions, enabling better staffing and resource allocation

### Cost Reduction

Prevent unnecessary readmissions saving an estimated 15-25% of readmission costs through targeted interventions

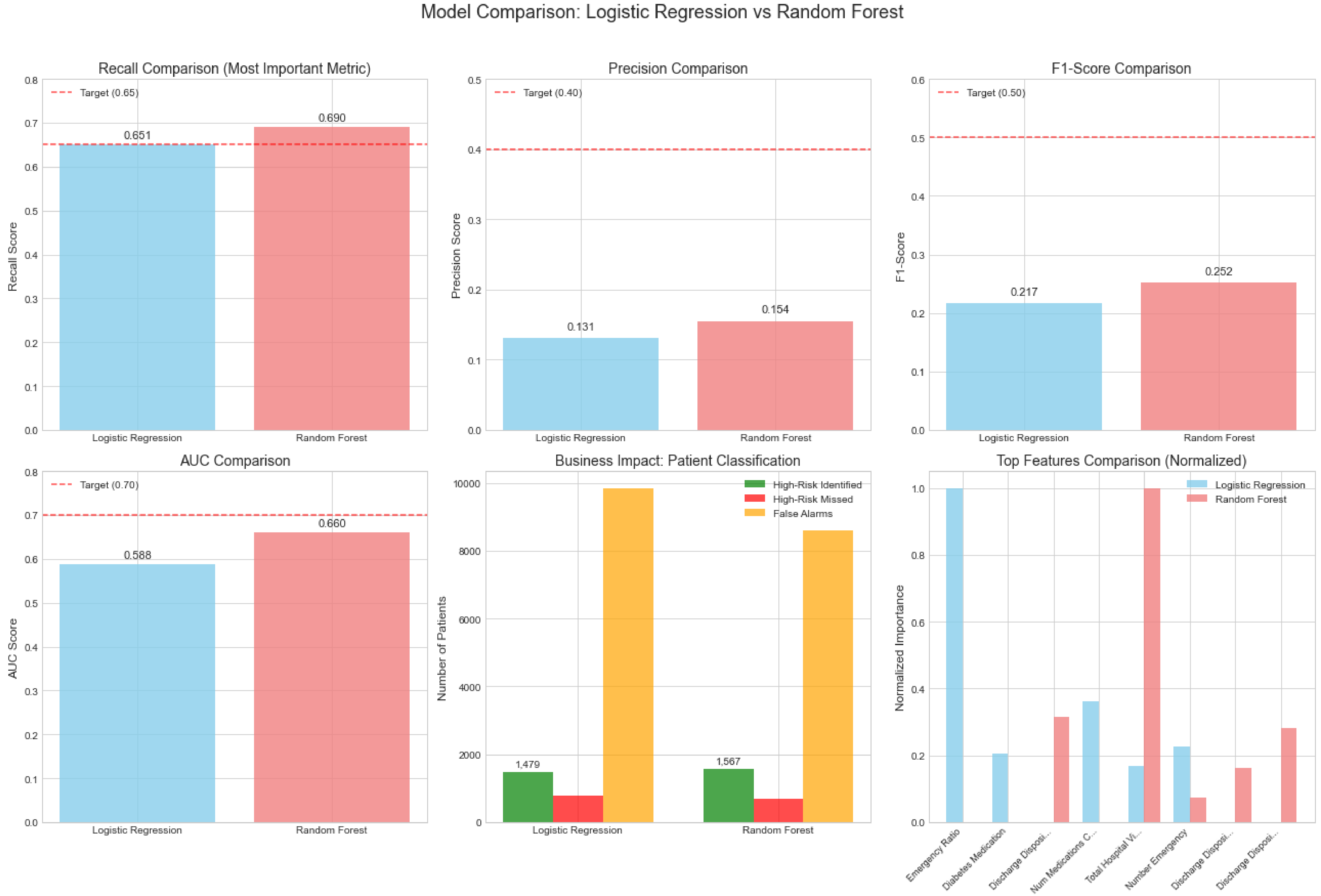
For a 500-bed hospital, preventing just 10% of readmissions could save $500,000-$1M annually


Model Performance Heatmap Comparison
Performance Metrics
(Higher is Better)
Target: 0.65
Target: 0.4
Target: 0.5
Target: 0.7
0.651
0.131
0.217
0.588
0.690
0.154
0.252
0.660
0.655
0.163
0.261
0.672
Logistic Regression
Random Forest
XGBoost
Models
Recall
Precision
F1-Score
AUC
Metrics
Score
Business Impact Metrics
(Actual Values Shown)
1,479
9,851
1,567
8,615
1,487
7,645
High-Risk ID
False Alarms
Metrics
Normalized Score

## Model Improvement Summary: Logistic Regression

| Metric | Baseline | Improved | Target | Status |
|---|---|---|---|---|
| Recall | 0.480 | 0.651 | 0.650 | □ MET |
| Precision | 0.159 | 0.131 | 0.400 | □ |
| F1-Score | 0.238 | 0.217 | 0.500 | □ |
| AUC | 0.626 | 0.588 | 0.700 | □ |

• *Model identifies 65.1% of high-risk patients (Recall target: 65.0%)*
• *Critical for Kenya: Catches most patients needing intervention*
• *Trade-off: Low precision means many false alarms (resource consideration)*



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Number Emergency
Discharge Disposi...
Discharge Disposi...


# Implementation Roadmap

## Path Forward

**Successful deployment requires collaboration between data science teams, clinicians, and hospital administrators to integrate predictions into existing workflows.**

### System Integration

**Integrate model into electronic health records so risk scores appear automatically at discharge planning.**

### Clinician Training

**Train healthcare providers on interpreting risk scores and implementing appropriate interventions for flagged patients.**

### Monitor & Refine

**Continuously track outcomes and retrain model with new data to improve accuracy over time.**

# Thank You

**Questions and Discussion - We welcome your feedback and insights**